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As certain as debt and taxes: Estimating the tax sensitivity of leverage from
state tax changes

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1 Research question

How sensitive is corporate leverage to changes in corporate income taxes, and do firms respond symmetrically to tax increases and tax cuts?

Answer: Using staggered changes in U.S. *state* corporate income tax rates as quasi-exogenous shocks, the paper finds that firms *increase* long-term leverage after tax *increases* (roughly +1 percentage point in long-term book leverage one year later), but show little or no deleveraging after tax *cuts*. This asymmetric response is consistent with *dynamic* tradeoff models featuring leverage ratcheting / hysteresis (rather than a frictionless static tradeoff model that predicts symmetry).

2 Why taxes identify leverage choices: the theory backbone

The paper is grounded in the classic idea that debt has *tax benefits* (interest deductibility) but also *costs* (financial distress, agency costs, adjustment costs). Tax changes provide a clean shifter of the marginal benefit of debt.

2.1 Static tradeoff theory: symmetric comparative statics

In a static tradeoff model, the firm chooses a debt level D to maximize value:

$$V(D) = \underbrace{\text{TaxBenefit}(D; \tau_c)}_{\uparrow \text{ in } \tau_c} - \underbrace{\text{Cost}(D)}_{\uparrow \text{ in } D}. \quad (1)$$

The optimal D^* satisfies the first-order condition

$$MB(D^*; \tau_c) = MC(D^*), \quad (2)$$

where MB is the marginal tax benefit and MC is the marginal cost of debt.

Static prediction: If a tax increase raises the marginal tax benefit, the optimal D^* rises. A tax cut of the same size lowers the marginal tax benefit and implies a *mirror-image* decrease in D^* . Hence leverage should respond *symmetrically* to increases vs. cuts.

2.2 Dynamic tradeoff theory: adjustment frictions and the “ratchet”

Dynamic models add frictions that can make leverage *stickier on the downside*:

- **Adjustment costs** for changing capital structure (issuance/retirement costs, renegotiation, market timing constraints).
- **Debt overhang / agency conflicts** that make paying down debt less attractive to equity holders.
- **Discrete refinancing opportunities** (capital structure moves on lumpy events rather than continuously).

These mechanisms can yield a *ratchet effect*: firms lever up when the tax benefit rises, but do not undo leverage when the tax benefit falls.

Figure 1: provides the intuition graphically by contrasting a symmetric static tradeoff with a dynamic/ratcheting framework.

2.3 A simple comparative-static expression (from the paper’s discussion)

Following the Miller (1977) logic, a textbook expression for the *tax advantage of debt* can be written as

$$\text{TaxAdvantage}(\tau_i, \tau_c, \tau_e) \cdot D = \left[(1 - \tau_i) - (1 - \tau_c)(1 - \tau_e) \right] D, \quad (3)$$

where τ_i is the personal tax rate on interest income, τ_e is the effective personal tax rate on equity income, and τ_c is the corporate tax rate.

To see why a higher equity-income tax τ_e dampens the response of debt to corporate tax changes, consider a simple quadratic cost of debt:

$$\text{Cost}(D) = a + bD + cD^2, \quad c > 0. \quad (4)$$

The FOC implies that the sensitivity of the optimal debt choice D^* with respect to τ_c is positive and proportional to $(1 - \tau_e)$:

$$\frac{\partial D^*}{\partial \tau_c} = \frac{1 - \tau_e}{2c} > 0. \quad (5)$$

Empirical implication: firms facing lower τ_e should react more strongly to corporate tax shocks (Table 8 exploits this using dividend-paying status as a proxy for τ_e).

3 Institutional setting and variation: state corporate income tax changes

3.1 What varies and why it is useful

The identification comes from top marginal corporate income tax rate changes at the *state* level. The paper exploits staggered timing and cross-state heterogeneity in these shocks.

3.2 Counts and magnitudes of tax shocks

Over the sample used for firm outcomes (fiscal years 1989–2011), the paper documents:

- **43** state corporate tax **increases** in **24** states affecting **2,212** sample firms.
- **78** state corporate tax **cuts** in **27** states affecting **7,283** sample firms.

Tax increases are economically meaningful: the *average* tax shock increases rates by about 93 basis points, while the average treated firm experiences a larger increase (about 131 basis points) because more firms are located in states with bigger increases. Tax cuts average about 55 basis points (about 69 basis points for the average sample firm due to firm geography).

3.3 How endogenous are tax changes? A state-level “first-stage” model (Table 1)

Although the core analysis treats state tax changes as quasi-exogenous for firms, the paper explicitly studies *which state conditions predict tax changes*. This serves two purposes: (i) it clarifies the political-economy origins of tax changes, and (ii) it motivates later controls for coincident policies.

A stylized representation of the models behind Table 1 is:

$$\text{TaxRise}_{s,t} = \alpha_s + \lambda_t + \psi'W_{s,t-1} + u_{s,t}, \quad (6)$$

$$\text{TaxCut}_{s,t} = \alpha_s + \lambda_t + \tilde{\psi}'W_{s,t-1} + \tilde{u}_{s,t}, \quad (7)$$

$$\Delta T_{s,t} = \alpha_s + \lambda_t + \kappa'W_{s,t-1} + \nu_{s,t}, \quad (8)$$

where $W_{s,t-1}$ includes political variables (election cycle, governor party), fiscal conditions (budget balance/deficit/surplus), credit ratings, and “tax competition” (relative rates vs neighbors). The empirical takeaway is that tax changes are predictable from political and fiscal state conditions, but these predictors are not obvious firm-level leverage shocks once one controls for industry-year conditions and firm fundamentals.

3.4 Figure 2: geography and “clustering” check

Figure 2 maps treated states by five-year periods to show that tax changes are not concentrated in a small region or a short subperiod. This supports the plausibility that tax changes are not simply a regional business-cycle phenomenon.

4 Data and measurement

4.1 Firm sample and outcome variables

The main firm sample is U.S. publicly traded, nonfinancial and nonutility firms (CRSP–Compustat), 1989–2011, totaling 91,487 firm-years.

Leverage outcomes: the baseline dependent variable is the *change in long-term book leverage*, defined as long-term debt divided by book assets (see Appendix C of the paper for exact Compustat items). The paper also studies alternative leverage measures and debt levels.

4.2 Key measurement issue: headquarter location

A crucial practical challenge is that Compustat reports a firm’s *current* headquarters address, which may differ from the *historical* headquarters relevant for earlier tax exposures. This creates misclassification:

- **False negatives:** firms that were in a tax-changing state at the time but appear not to be.
- **False positives:** firms that appear exposed to a tax change but were not.

This attenuates estimated tax sensitivity toward zero. The paper therefore constructs a backfilled headquarters history using SEC filings (EDGAR) and related sources. Table 3 shows that cor-

recting measurement error *increases* the estimated effect (consistent with attenuation bias under misclassification).

5 Empirical strategy: baseline diff-in-diff in first differences

5.1 Baseline specification (Equation 1)

Let i index firms, j industries, s states, and t years. Let D_{ijst} be a debt usage measure (baseline: long-term book leverage). The core regression is estimated in *first differences*:

$$\Delta D_{ijst} = \beta \Delta T_{s,t-1}^+ + \gamma \Delta T_{s,t-1}^- + \delta' \Delta X_{i,t-1} + \theta' \Delta Z_{s,t-1} + \alpha_{j,t} + \varepsilon_{ijst}, \quad (9)$$

where:

- Δ is the first-difference operator.
- $\Delta T_{s,t-1}^+$ equals 1 if state s *increased* its top marginal corporate tax rate in year $t - 1$.
- $\Delta T_{s,t-1}^-$ equals 1 if state s *cut* its top marginal corporate tax rate in year $t - 1$.
- $X_{i,t-1}$ are time-varying firm controls (profitability, size, tangibility, market-to-book, etc.).
- $Z_{s,t-1}$ are state controls (growth, unemployment, etc.).
- $\alpha_{j,t}$ are industry-year fixed effects ($\text{SIC4} \times \text{year}$), absorbing time-varying industry shocks.
- Standard errors are clustered at the state level.

5.2 Why first differences?

First differencing removes firm fixed effects from the corresponding levels equation and naturally accommodates repeated treatments (a firm can experience multiple tax changes over time).

5.3 Identification assumption: parallel trends

The causal interpretation requires that absent a tax change, treated firms would have followed the same trend in leverage as control firms (conditional on controls and industry-year effects).

The paper evaluates this assumption in multiple ways:

1. **Graphical event study:** Figure 3 shows no meaningful pre-trends before tax changes and a sharp post-tax-increase response with a one-year lag.
2. **Leads/lags regression:** Table 4 estimates leads and lags of tax indicators.
3. **Falsification and border designs:** Table 6 uses neighbor-state tax changes and contiguous border counties.
4. **Confounds checks:** Table 7 controls for other state policy changes and partitions by union power and unemployment risk.

5.4 Event-study / leads-and-lags specification (Table 4)

A convenient way to formalize Figure 3 and Table 4 is to replace the single lagged treatment indicator with a set of leads and lags relative to the tax-change year:

$$\Delta D_{ijst} = \sum_{k=-K}^K \beta_k \text{TaxRise}_{s,t-k} + \sum_{k=-K}^K \gamma_k \text{TaxCut}_{s,t-k} + \delta' \Delta X_{i,t-1} + \theta' \Delta Z_{s,t-1} + \alpha_{j,t} + \varepsilon_{ijst}, \quad (10)$$

with one event-time indicator omitted as the reference period. The identifying “parallel trends” restriction is that the *lead* coefficients (pre-treatment $k < 0$) are close to zero.

5.5 Cross-sectional persistence / hysteresis in leverage levels

To test whether tax *increases* leave a persistent imprint on leverage (while tax cuts do not), the paper estimates a cross-sectional regression in levels using each firm’s last observed year τ :

$$D_{i,\tau} = \phi + \beta N_{i,\tau}^+ + \gamma N_{i,\tau}^- + \xi' X_{i,\tau} + \alpha_{j,\tau} + \varepsilon_i, \quad (11)$$

where $N_{i,\tau}^+$ ($N_{i,\tau}^-$) counts the number of tax increases (cuts) the firm has experienced since entering the panel. A positive β and near-zero γ is interpreted as evidence of hysteresis/ratcheting.

5.6 Neighbor-state falsification and border-county design (Table 6)

To address the concern that tax changes proxy for local business-cycle shocks, the paper uses two related designs.

Neighbor-state falsification. Augment the baseline model with tax changes in *bordering* states:

$$\Delta D_{ijst} = \beta \text{TaxRise}_{s,t-1} + \gamma \text{TaxCut}_{s,t-1} + \beta^N \text{NeighborRise}_{s,t-1} + \gamma^N \text{NeighborCut}_{s,t-1} + \dots \quad (12)$$

If the estimated tax effect were really driven by regional shocks, one might expect the neighbor coefficients to mimic the own-state coefficient. They do not.

Contiguous border counties. Restrict the sample to firms headquartered in counties adjacent to a state border, and compare to firms in the contiguous county on the other side of that border. A stylized specification is:

$$\Delta D_{ijct} = \beta \text{TaxRise}_{s(c),t-1} + \gamma \text{TaxCut}_{s(c),t-1} + \delta' \Delta X_{i,t-1} + \alpha_{p(c),t} + \alpha_{j,t} + \varepsilon_{ijct}, \quad (13)$$

where c indexes counties, $s(c)$ is the state containing county c , and $\alpha_{p(c),t}$ is a county-pair-by-year fixed effect for the cross-border pair $p(c)$. This absorbs local economic conditions common to both sides of the border; identification comes from sharp tax-policy variation across the border.

6 Multi-state firms: weighted tax exposure (Equation 2)

Single-state headquarters exposure may be imperfect for multi-state firms. The paper therefore builds a *firm-specific weighted tax shock* using establishment-level information from NETS (National

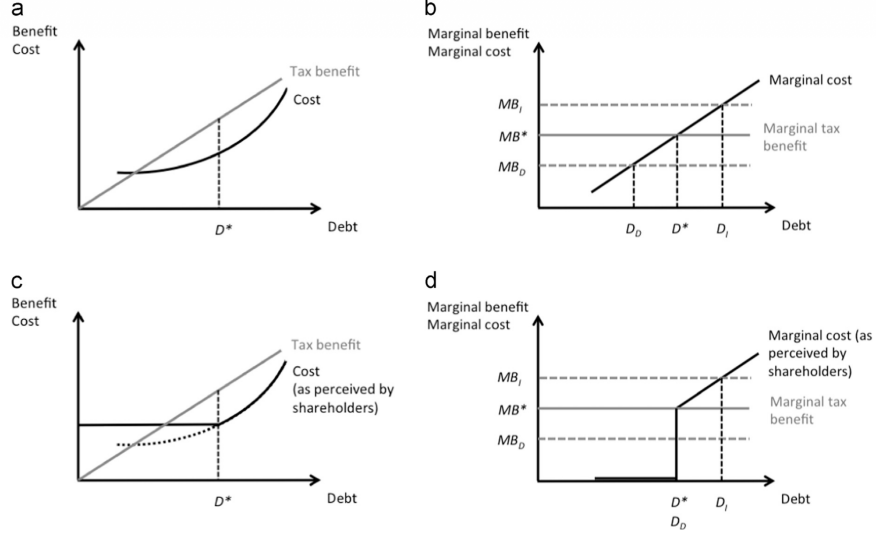


Fig. 1. Illustrating static and dynamic tradeoff theory. In static tradeoff theory, shown in (a), the optimal level of debt D^* maximizes firm value by

Establishment Time Series):

$$\Delta\tau_{i,t} = \sum_s \left[\frac{1}{2} \frac{\text{employees}_{i,s,t}}{\text{employees}_{i,\text{total},t}} + \frac{1}{2} \frac{\text{sales}_{i,s,t}}{\text{sales}_{i,\text{total},t}} \right] \Delta T_{s,t}. \quad (14)$$

This approximates the firm's nexus with each state using a 50/50 average of employment and sales shares.

7 Interpreting the figures

7.1 Figure 1: static vs. dynamic tradeoff

Interpretation: The figure is not a result; it is the conceptual map.

- In the **static** model, tax increases and tax cuts shift the marginal benefit curve up/down and should induce *symmetric* leverage adjustments.
- In the **dynamic** model, leverage can *ratchet*: tax increases raise debt, but tax cuts do not induce commensurate deleveraging due to adjustment frictions or agency problems.

This is exactly what the data later show.

7.2 Figure 2: where and when tax shocks occur

Interpretation: State tax changes occur throughout the sample window and are geographically dispersed. The absence of strong clustering makes it less plausible that the leverage response is driven by a single regional recession or localized policy package.

7.3 Figure 3: event-study evidence for timing and asymmetry

Figure 3 plots within-firm changes in long-term leverage for treated firms vs. control firms around the event year.

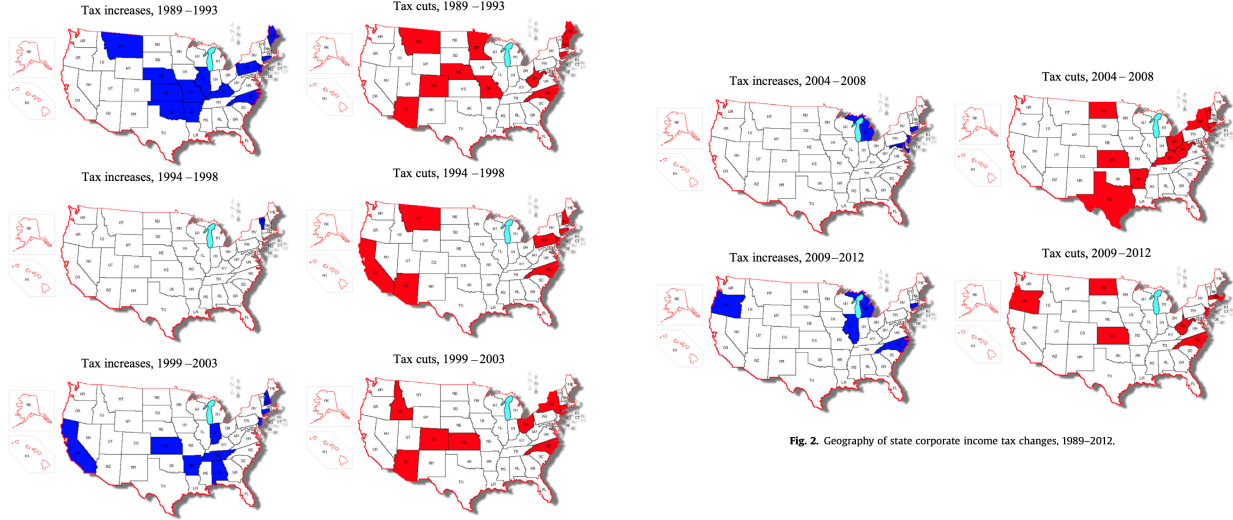


Fig. 2. Geography of state corporate income tax changes, 1989–2012.

- **Tax increases:** pre-trends are near zero; leverage rises sharply *one year after* the tax increase.
- **Tax cuts:** there is no comparably strong deleveraging response.

This supports (i) the identifying parallel-trends assumption and (ii) the dynamic/ratchet interpretation.

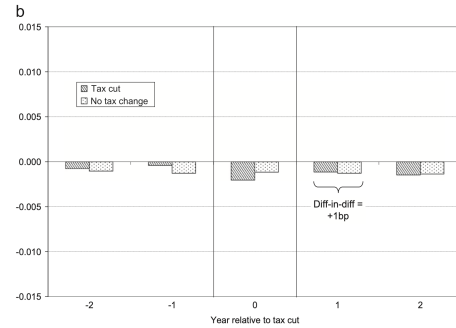
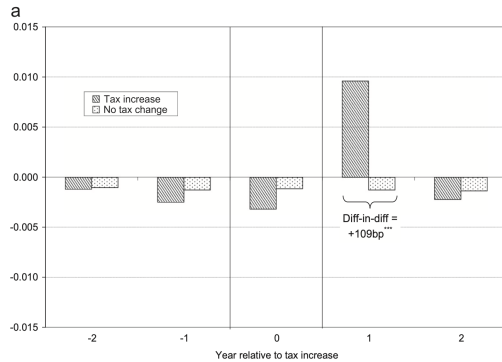


Fig. 3. Annual changes in leverage around state tax increases and state tax cuts. The figures plot the average annual within-firm change

8 Interpreting the tables

8.1 Table 1: determinants of tax changes and coincident state-level policy changes

What it does: Table 1 asks whether tax changes are systematically related to observable political/economic state conditions (and whether other state policies shift at the same time).

Panel A (political and economic correlates):

- Tax increases are more likely in states with certain political-economy features (e.g., election timing, party affiliation) and fiscal conditions (budget deficits), and are related to *tax competition* with neighboring states.

Table 1

Determinants of state corporate income tax changes and coincident state-level changes. Panel A models the determinants of the probability that a state changes its top marginal corporate income tax rate and the magnitude of any such change. Columns 1–3 report summary statistics of the explanatory variables, showing fractions or means (with standard deviations shown in italics underneath the means). Column 4 compares conditions in states that increase taxes to those in states that cut taxes. Columns 5 and 6 model the probability that a state raises or cuts corporate income taxes, using linear probability models. Columns 7 to 9 model the magnitude of the tax changes (measured in percentage points). All specifications are estimated using least squares with state and year fixed effects (not shown for brevity). The sample covers 50 U.S. states plus DC in 1990–2011, for a total of 1,122 observations. In columns 5–9, heteroskedasticity-consistent standard errors clustered at the state level are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 1%, 5%, and 10% level (two-sided), respectively. Panel B reports state-level changes in economic quantities that coincide with either increases or cuts in state corporate income taxes and that have a plausible basis in theory to potentially affect corporate leverage decisions. We focus on changes in state personal income tax rates, state capital gains tax rates, or state taxes on banks and changes in state investment incentive programs (i.e., tax credits for investment, R&D, and job creation, as well as job creation grant programs). For variable definitions and details of their construction, see [Appendix C](#).

	Summary statistics			Difference	Probability of...		Magnitude of...		
	All obs. (1)	Tax increases (2)	Tax cuts (3)	(tax inc. – tax cut) (4)	Tax increase (5)	Tax cut (6)	Tax change (7)	Tax increase (8)	Tax cut (9)
Panel A									
Political conditions									
= 1 if Democratic governor in year $t-1$	0.472	0.622	0.392	0.230**	0.007	–0.037**	0.067**	0.034	–0.033**
= 1 if 1 year to next gubernatorial election					0.014	0.017	0.026	0.021	0.014
= 1 if 2 years to next gubernatorial election					0.013	0.060**	0.024	0.028	0.005
= 1 if 3 years to next gubernatorial election					0.018	0.030	0.036	0.017	0.032
					–0.003	0.026	0.029	0.027	–0.005
					0.016	0.025	0.040	0.021	0.035
					0.053**	0.044	0.040	0.044**	0.006
					0.024	0.029	0.035	0.020	0.028
Economic conditions (in year $t-1$)									
State budget balance	0.020	0.008	0.034	–0.027**			–0.408**		
	0.069	0.061	0.062				0.190		
State budget deficit	–0.014	–0.020			–0.665**			–1.023**	
	0.026	0.033			0.339			0.461	
State budget surplus	0.034		0.043			0.296*			0.093
	0.056		0.051			0.161			0.119
= 1 if state bond rating downgraded	0.044	0.135	0.000	0.135***	0.058	–0.059**	0.176	0.129	–0.042*
GSP growth rate	0.027	0.018	0.030	–0.012**	0.044	0.026	0.114	0.112	0.022
	0.028	0.023	0.027		0.248	0.336	–0.052	0.333	0.317
State unemployment rate	0.055	0.053	0.056	–0.002	0.231	0.336	0.473	0.299	0.353
	0.018	0.017	0.019		–0.251	1.000	–1.082	–0.249	0.752
State union penetration	0.085	0.092	0.083	0.009	0.601	1.106	1.080	0.849	0.568
	0.043	0.032	0.044		–0.368	–0.453	–0.349	–0.715	–0.229
					0.473	0.703	0.879	0.741	0.489
Tax competition (in year $t-1$)									
State's tax rate relative to highest tax rate among its neighboring states	–0.019	–0.020	–0.005	–0.015***	–0.021***	0.036***	–0.059***	–0.023**	0.036***
	0.037	0.028	0.020		0.007	0.010	0.016	0.012	0.010
Diagnostics					12.4%	21.7%	10.7%	11.0%	11.9%
R^2									
Panel B									
		Tax increases			Tax cuts				
Number of tax changes		43			78				
... of which coincide with									
	increase in state personal income tax rate	22			13				
	cut in state personal income tax rate	2			15				
	increase in state capital gains tax rate	21			12				
	cut in state capital gains tax rate	2			16				
	increase in state tax on banks	28			0				
	cut in state tax on banks	0			56				
	increase in state investment tax credit rate	1			6				
	cut in state investment tax credit rate	0			0				
	increase in state R&D credit rate	2			9				
	cut in state R&D credit rate	1			2				
	increase in state job creation credit	0			3				
	cut in state job creation credit	0			1				
	increase in state job creation grants	0			1				
	cut in state job creation grants	0			0				

- Importantly, many state-level correlates that matter for taxes do *not* have a direct structural link to firm leverage (e.g., party affiliation per se), which helps motivate the strategy: taxes shift for political reasons that are plausibly orthogonal to firm-level leverage shocks within industry-year cells.

Panel B (coincident policy changes):

- It documents how often corporate tax changes coincide with changes in other taxes (personal income, capital gains, bank taxes) and with investment incentive programs.
- This motivates later robustness checks: if other coincident changes drive leverage, then controlling for them should kill the corporate-tax coefficient. Table 7 shows they do not.

8.2 Table 2: firm-level summary statistics

Purpose: establishes baseline levels and dispersion for leverage and controls, and compares firms about to face a tax increase vs. a tax cut.

Key reading:

- Average long-term book leverage in the full sample is about 17%.

Table 2

Firm-level summary statistics. The sample consists of 91,487 firm-years for all nonfinancial and non-utility U.S. companies that are traded on the NYSE, Amex, or Nasdaq in fiscal years 1989 through 2011, as per the merged CRSP-Compustat Fundamentals Annual database. The table reports summary statistics for our dependent variables and the controls. For variable definitions and details of their construction, see [Appendix C](#). Return on assets, tangibility, firm size, and market/book are winsorized 0.5% in each tail.

	All firm-years ($N=91,487$)						One year before a tax increase		One year before a tax cut	
	Mean	S.D.	Percentile			First diff.	$(N=1,735)$		$(N=6,627)$	
			25th	50th	75th		Mean	S.D.	Mean	S.D.
Firm leverage										
Long-term book leverage	0.172	0.264	0.002	0.100	0.275	0.003	0.183	0.217	0.172	0.211
Long-term book lev. (incl. current portion)	0.198	0.295	0.006	0.133	0.310	0.006	0.208	0.228	0.194	0.222
Total book leverage	0.226	0.311	0.019	0.174	0.348	0.009	0.236	0.237	0.221	0.231
Long-term market leverage	0.176	0.222	0.001	0.079	0.281	0.008	0.199	0.230	0.173	0.219
Real long-term debt (\$m)	385.0	2,502.9	0.1	6.9	117.1	27.5	570.2	3,262.1	368.5	1,848.7
Firm characteristics										
ROA	0.034	0.273	0.009	0.104	0.166	-0.011	0.046	0.254	0.046	0.253
Firm size (total assets, \$m)	1,683.0	9,546.0	34.2	134.9	628.1	121.3	2,297.0	10,308.2	1,652.6	9,258.0
Tangibility	0.264	0.224	0.087	0.196	0.379	0.001	0.260	0.205	0.260	0.219
Market/book	1.840	1.939	0.813	1.210	2.054	-0.076	1.796	1.977	1.856	1.982
Credit market conditions										
Default spread (in %)	0.956	0.465	0.680	0.870	1.080	0.013	1.235	0.576	0.855	0.269
State characteristics										
GSP growth rate	0.029	0.026	0.012	0.029	0.046	-0.001	0.013	0.019	0.035	0.025
State unemployment rate	0.058	0.017	0.046	0.054	0.066	0.001	0.054	0.018	0.057	0.016

- Firms about to face a tax increase have somewhat higher leverage and are larger on average than firms about to face a tax cut, reinforcing the need for within-firm differencing and controls.

8.3 Table 3: baseline effect of tax changes on leverage (main result)

Column (1) baseline: a corporate tax *increase* is followed by a statistically and economically significant increase in long-term book leverage *one year later* (around +1 percentage point). A corporate tax *cut* shows no comparable effect.

How to read the columns (core logic):

1. **Measurement error check (col. 2):** using mismeasured HQ locations attenuates the effect, consistent with classical misclassification bias.
2. **Dose-response (col. 3–4):** larger tax increases produce larger leverage increases; using the *continuous* tax-rate change yields an economically interpretable slope (basis points of leverage per 1pp tax increase).
3. **Tax-base weighting (col. 5):** using weighted state exposures for multi-state firms yields similar estimates, suggesting that HQ-based treatment is not driving results mechanically.
4. **Pre-announced sequences (col. 6):** restricting tax-cut treatments to the first year of a multi-year cut does not uncover a hidden response to tax cuts.
5. **Within-firm robustness (col. 7):** adding firm fixed effects to the first-differenced equation leaves conclusions unchanged.
6. **Reversals / hysteresis (col. 8):** firms that experience an increase and later a cut still do not reduce leverage after the cut, supporting hysteresis.
7. **Debt levels (col. 9):** using log real long-term debt confirms that the leverage response is not purely denominator-driven.

Table 3

Effect of tax changes on leverage. We estimate standard leverage regressions to test whether, and by how much, firms change their leverage in response to changes in U.S. state corporate income taxes in their headquarter state. For variable definitions and details of their construction, see Appendix C. For a description of the sample, see Table 2. Except in columns 4 and 5, we capture tax changes using indicator variables for tax increases and tax cuts. In columns 4 and 5, we use changes in a state's top marginal corporate income tax rate. Note that two corporate income tax increases (CA 2002 and NJ 2002) and one tax cut (TX 2008) cannot be summarized in terms of changes in marginal tax rates; see Appendices A and B. In column 3, large, medium, and small tax changes are those in the top, middle, and bottom terciles. Column 5 uses the weighted average change in tax rates as defined in Eq. (2). Column 6 restricts the tax-cut treatment to the first in a pre-announced sequence of annual tax cuts. (For example, NY cut the top corporate income tax rate from 9% to 7.5% in three annual half-percentage-point increments starting in 1999.) Column 8 restricts the sample of treated firms to those that suffer first a tax increase and then a subsequent tax cut ("reversals"). Column 9 models log real debt rather than a leverage ratio. The unit of analysis in each column is a firm-year. All specifications are estimated using ordinary least squares (OLS) in first differences to remove firm fixed effects (FE) in the levels equations and include industry-year fixed effects to remove industry shocks. The specifications shown in columns 7 and 8 additionally include firm fixed effects in the first-difference equation and are estimated using Stata's *reg2hdfe* command for linear regressions with two high-dimensional fixed effects. The fixed effects are not reported for brevity. Heteroskedasticity-consistent standard errors clustered at the state level are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 1%, 5%, and 10% level (two-sided), respectively.

	Dep. var.: Change in long-term book leverage								Dep. var.: Change in log real long-term debt
	Baseline (1)	Baseline w/backfilled Compustat HQ data (2)	Large vs. small tax changes (3)	Baseline w/changes in marginal tax rates (4)	Baseline w/weighted changes in tax rates (5)	Baseline w/o late-in- sequence tax cuts (6)	Baseline w/firm FE (7)	Reversals w/firm FE (8)	
Tax increase indicators (in %, except col. 9) = 1 if tax rise at $t-1$	1.042*** 0.196	0.746*** 0.204	1.310*** 0.406			1.030*** 0.194	1.161*** 0.215	1.064*** 0.307	0.057** 0.025
= 1 if large tax rise at $t-1$				0.907*					
= 1 if medium tax rise at $t-1$				0.522					
= 1 if small tax rise at $t-1$				0.705					
				0.525					
Tax cut indicators (in %, except col. 9) = 1 if tax cut at $t-1$	-0.010 0.136	-0.045 0.167				-0.153 0.131	-0.097 0.200	-0.071 0.487	-0.003 0.867
= 1 if large tax cut at $t-1$			0.009						
= 1 if medium tax cut at $t-1$			0.239						
= 1 if small tax cut at $t-1$			0.160						
			0.347						
			-0.112						
			0.246						
Continuous tax changes Size of corporate tax rise at $t-1$				0.387**	0.405**				
				0.168	0.176				
Size of corporate tax cut at $t-1$				0.168	0.235				
				0.128					
Lagged change in...									
ROA	-0.005	-0.005	-0.005	-0.004	-0.004	-0.005	-0.001	0.001	-0.015
	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.006	0.021
Firm size	0.007***	0.007***	0.007***	0.007***	0.007***	0.007***	0.003	0.002	0.149***
	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.004	0.014
Tangibility	0.037***	0.037***	0.037***	0.037***	0.037***	0.037***	0.026**	0.025	0.389***
	0.011	0.011	0.009	0.011	0.011	0.011	0.011	0.008	0.071
Market/book	0.000	0.000	0.000	0.000	0.000	0.000	-0.001	-0.001	0.012***
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.003
Default spread	-0.512***	-0.512***	-0.512***	-0.516***	-0.536***	-0.514***	-0.507***	-0.502***	-0.057***
	0.123	0.124	0.168	0.127	0.127	0.123	0.123	0.216	0.012
GSP growth rate	0.025	0.025	0.025	0.025	0.025	0.024	0.020	-0.010	0.189
	0.019	0.019	0.023	0.019	0.019	0.019	0.020	0.042	0.190
State unemployment rate	0.171**	0.176**	0.173*	0.168*	0.174**	0.162*	0.087	0.039	1.308**
	0.084	0.086	0.096	0.084	0.086	0.085	0.104	0.157	0.603
Diagnostics									
R ²	11.28	11.28	11.28	11.28	11.28	11.28	21.48	31.88	11.48
Wald test: all coeff. = 0	13.4***	9.6***	14.8***	11.6***	9.6***	13.4***	n.a.	n.a.	28.7***
No. of firms	8,867	8,867	8,867	8,864	8,864	8,867	8,867	5,456	8,867
No. of observations	73,832	73,832	73,832	73,228	73,228	73,832	73,832	36,087	73,832

8.4 Table 4: reversals, delayed reactions, and pre-trends

Goal: test dynamic timing and whether trends differ before treatment.

Main message:

- The only robustly significant coefficient for tax increases is at the one-year lag (consistent with Figure 3).
- Leads are small and insignificant: no evidence firms increase leverage *before* tax increases or cuts, supporting parallel trends and arguing against reverse causality.
- Post-treatment lags do not show meaningful reversal: the leverage increase is persistent, consistent with ratcheting.

8.5 Table 5: robustness across samples and alternative outcomes

What changes: splits by time period, adds state fixed effects, changes leverage definitions, and studies extensive margins.

Reading:

- The positive response to tax *increases* is stable pre- and post-2000 and survives adding state fixed effects.

Table 4

Testing for reversals and pre-trends. To investigate possible reversals and pre-trends, we include four lags and up to one lead in the baseline regression shown in Table 3, column 1. For variable definitions and details of their construction, see Appendix C. For a description of the sample, see Table 2. The unit of analysis is a firm-year. All specifications are estimated using OLS in first differences to remove firm fixed effects in the levels equations and include industry-year fixed effects to remove industry shocks. The fixed effects are not reported for brevity. Heteroskedasticity-consistent standard errors clustered at the state level are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 1%, 5%, and 10% level (two-sided), respectively.

		Dep. var.: Change in long-term book leverage		
		(1)	(2)	(3)
Tax increase	at $t+1$ (in %)	-0.121 <i>0.302</i>		
	at $t=0$ (in %)	-0.234 <i>0.214</i>	-0.261 <i>0.232</i>	-0.369 <i>0.241</i>
	at $t=-1$ (in %)	1.172*** <i>0.266</i>	1.229*** <i>0.209</i>	0.962*** <i>0.195</i>
	at $t=-2$ (in %)	-0.328 <i>0.282</i>	-0.335 <i>0.285</i>	-0.392 <i>0.263</i>
	at $t=-3$ (in %)	-0.126 <i>0.233</i>	0.000 <i>0.232</i>	0.088 <i>0.221</i>
	at $t=-4$ (in %)	0.159 <i>0.362</i>	0.433 <i>0.315</i>	
	at $t+1$ (in %)	0.137 <i>0.171</i>		
	at $t=0$ (in %)	0.142 <i>0.172</i>	-0.003 <i>0.174</i>	0.053 <i>0.175</i>
Tax cut	at $t=-1$ (in %)	-0.170 <i>0.188</i>	-0.019 <i>0.186</i>	-0.094 <i>0.197</i>
	at $t=-2$ (in %)	-0.195 <i>0.210</i>	-0.028 <i>0.170</i>	0.003 <i>0.165</i>
	at $t=-3$ (in %)	-0.032 <i>0.146</i>	-0.008 <i>0.162</i>	0.024 <i>0.187</i>
	at $t=-4$ (in %)	0.184 <i>0.125</i>	0.116 <i>0.149</i>	
	at $t+1$ (in %)	0.137 <i>0.171</i>		
	at $t=0$ (in %)	0.142 <i>0.172</i>	-0.003 <i>0.174</i>	0.053 <i>0.175</i>
	at $t=-1$ (in %)	-0.170 <i>0.188</i>	-0.019 <i>0.186</i>	-0.094 <i>0.197</i>
	at $t=-2$ (in %)	-0.195 <i>0.210</i>	-0.028 <i>0.170</i>	0.003 <i>0.165</i>
Lagged change in...	ROA	-0.003 <i>0.006</i>	-0.002 <i>0.004</i>	-0.004 <i>0.005</i>
	Firm size	0.008*** <i>0.002</i>	0.006*** <i>0.002</i>	0.007*** <i>0.002</i>
	Tangibility	0.035*** <i>0.013</i>	0.036*** <i>0.013</i>	0.034*** <i>0.011</i>
	Market/book	0.001* <i>0.0005</i>	0.001 <i>0.0005</i>	0.000 <i>0.0005</i>
	Default spread	-0.430*** <i>0.122</i>	-0.468*** <i>0.107</i>	-0.483*** <i>0.104</i>
	GSP growth rate	0.005 <i>0.025</i>	0.003 <i>0.023</i>	0.015 <i>0.020</i>
	State unemployment rate	0.142 <i>0.118</i>	0.110 <i>0.104</i>	0.133 <i>0.107</i>
	Diagnostics			
Diagnostics	R^2	14.2%	13.2%	12.0%
	Wald test: all coeff.=0	7.3***	8.1***	9.3***
	No. of firms	6,582	7,375	8,192
	No. of observations	54,318	61,374	69,203

Table 5

Robustness. To investigate robustness, columns 1 and 2 split the sample in 2000; column 3 adds state fixed effects; column 4 models total leverage; column 5 models long-term leverage including debt due within one year; column 6 models market leverage; and columns 7 and 8 model the probability that the firm issues or repurchases long-term debt, respectively. For variable definitions and details of their construction, see Appendix C. For a description of the sample, see Table 2. The unit of analysis is a firm-year. All specifications are estimated using OLS in first differences to remove firm fixed effects in the levels equations and include industry-year fixed effects to remove industry shocks. (The specification shown in column 3 additionally includes state fixed effects and is estimated using Stata's `reg2hdfe` command for linear regressions with two high-dimensional fixed effects.) The fixed effects are not reported for brevity. Heteroskedasticity-consistent standard errors clustered at the state level are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 1%, 5%, and 10% level (two-sided), respectively.

	Dep. var.: Change in book leverage					Dep. var.:		
	Long-term debt, 1990-2000 (1)	Long-term debt, 2001-2011 (2)	Long-term debt (w/state FE) (3)	Total (short- and long-term) debt (4)	Long-term (incl. current portion) (5)	Change in long-term market leverage (6)	Prob(issue long-term debt) (7)	Prob(debt repurchase) (8)
= 1 if tax rise at $t=-1$ (in %, except cols. 7 & 8)	1.070*** <i>0.322</i>	1.025*** <i>0.236</i>	1.038*** <i>0.225</i>	0.830*** <i>0.225</i>	1.118*** <i>0.191</i>	0.660* <i>0.342</i>	0.038*** <i>0.010</i>	-0.017* <i>0.010</i>
= 1 if tax cut at $t=-1$ (in %, except cols. 7 & 8)	-0.103 <i>0.188</i>	0.158 <i>0.171</i>	-0.007 <i>0.190</i>	-0.032 <i>0.194</i>	-0.013 <i>0.131</i>	0.069 <i>0.174</i>	0.004 <i>0.785</i>	0.001 <i>1.187</i>
Lagged change in ...								
ROA	-0.001 <i>0.006</i>	-0.009** <i>0.003</i>	-0.004 <i>0.004</i>	-0.010*** <i>0.004</i>	-0.012*** <i>0.004</i>	-0.017*** <i>0.003</i>	-0.031*** <i>0.009</i>	0.046*** <i>0.010</i>
Firm size	0.006** <i>0.002</i>	0.007*** <i>0.003</i>	0.007*** <i>0.002</i>	0.011*** <i>0.002</i>	0.012*** <i>0.002</i>	0.027*** <i>0.003</i>	0.112*** <i>0.008</i>	-0.065*** <i>0.007</i>
Tangibility	0.038*** <i>0.013</i>	0.034*** <i>0.016</i>	0.037*** <i>0.011</i>	0.063*** <i>0.013</i>	0.057*** <i>0.013</i>	0.060*** <i>0.010</i>	0.258*** <i>0.033</i>	-0.164*** <i>0.040</i>
Market/book	0.000 <i>0.001</i>	-0.001*** <i>0.0004</i>	0.000 <i>0.000</i>	-0.001** <i>0.0004</i>	-0.001 <i>0.000</i>	0.001*** <i>0.0004</i>	0.007*** <i>0.002</i>	-0.005*** <i>0.001</i>
Default spread	-0.188 <i>0.502</i>	-0.579*** <i>0.129</i>	-0.512*** <i>0.123</i>	-0.671*** <i>0.148</i>	-0.551*** <i>0.126</i>	-1.345*** <i>0.168</i>	-0.029*** <i>0.007</i>	0.038*** <i>0.007</i>
GSP growth rate	0.039 <i>0.027</i>	0.014 <i>0.022</i>	0.024 <i>0.019</i>	0.033* <i>0.017</i>	0.036** <i>0.016</i>	0.017 <i>0.023</i>	0.032 <i>0.089</i>	-0.049 <i>0.088</i>
State unemployment rate	0.170 <i>0.111</i>	0.186 <i>0.128</i>	0.138* <i>0.084</i>	0.113 <i>0.090</i>	0.175** <i>0.085</i>	0.166 <i>0.107</i>	-0.113 <i>0.334</i>	-0.022 <i>0.606</i>
Diagnostics								
R^2	10.7%	12.0%	11.2%	11.3%	11.1%	20.3%	16.4%	14.0%
Wald test: all coeff.=0	6.7***	13.2***	n.a.	52.0***	24.2***	49.1***	45.9***	38.6***
No. of firms	7,237	5,085	8,867	8,840	8,851	8,862	8,867	8,867
No. of observations	40,923	32,909	73,832	73,544	73,673	73,777	73,832	73,832

Table 6

Potential confound: local business cycle effects. States may change corporate tax rates, and firms may change leverage, in response to unobserved changes in local business conditions. To examine this potential confound, column 1 estimates a falsification test, asking if firms respond to tax changes that occur in a neighboring state when not themselves experiencing a tax change in their own home state. Columns 2 and 3 restrict the set of control firms to those located in a neighboring state, thus excluding far-away states. Column 2 focuses on firms in a tax-increase state and their untreated neighbors, while column 3 focuses on firms in a tax-cut state and their untreated neighbors; firms in states that neither experience a tax change nor border a state that does are excluded. Column 4 uses a restricted sample consisting of firms in contiguous counties on either side of a state border, such that in year t , one or more firms in one county experience a tax shock while one or more firms in the contiguous county do not. The effect of common local economic shocks are then removed by including county-pair/year fixed effects. Column 5 additionally requires that firms in contiguous county-pairs operate in the same SIC4 industry in year t . The unit of analysis is a firm-year. All specifications except column 5 are estimated using OLS in first differences with industry-year fixed effects (not shown for brevity). Column 5 instead includes county-pair/industry/year fixed effects. For variable definitions and details of their construction, see [Appendix C](#). For a description of the sample, see [Table 2](#). Heteroskedasticity-consistent standard errors clustered at the state level are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 1%, 5%, and 10% level (two-sided), respectively.

	Dep. var.: Change in long-term book leverage				
	Full sample (1)	Firms located in		Firms in contiguous border counties	
		Tax-increase states and their neighbors (2)	Tax-cut states and their neighbors (3)	County-group/year FE & industry/year FE (4)	County-group/ industry/year FE (5)
= 1 if tax rise at $t = -1$ (in %)	0.931*** <i>0.201</i>	1.490*** <i>0.321</i>		1.305** <i>0.666</i>	1.280*** <i>0.382</i>
= 1 if tax cut at $t = -1$ (in %)	0.011 <i>0.124</i>		-0.056 <i>0.185</i>	0.045 <i>0.582</i>	-0.109 <i>0.597</i>
= 1 if tax rise in a bordering state at $t = -1$ (in %)	-0.417*** <i>0.142</i>				
= 1 if tax cut in a bordering state at $t = -1$ (in %)	0.067 <i>0.081</i>				
Lagged change in ...					
ROA	-0.004 <i>0.004</i>	-0.019* <i>0.010</i>	-0.005 <i>0.005</i>	0.003 <i>0.012</i>	0.008 <i>0.022</i>
Firm size	0.007*** <i>0.002</i>	0.010** <i>0.004</i>	0.006 <i>0.004</i>	0.008 <i>0.006</i>	-0.004 <i>0.009</i>
Tangibility	0.037*** <i>0.011</i>	-0.001 <i>0.030</i>	0.054*** <i>0.015</i>	0.056 <i>0.035</i>	-0.071 <i>0.043</i>
Market/book	0.000 <i>0.000</i>	-0.002* <i>0.001</i>	-0.001* <i>0.0006</i>	0.001 <i>0.001</i>	0.002 <i>0.001</i>
Default spread	-0.524*** <i>0.124</i>	-0.755** <i>0.293</i>	-0.780** <i>0.312</i>	-1.419** <i>0.673</i>	-0.447 <i>1.102</i>
GSP growth rate	0.025 <i>0.019</i>	0.184** <i>0.079</i>	0.021 <i>0.038</i>	0.127 <i>0.182</i>	-0.051 <i>0.085</i>
State unemployment rate	0.198** <i>0.077</i>	0.408** <i>0.167</i>	0.367 <i>0.245</i>	-0.007 <i>0.009</i>	-0.014*** <i>0.005</i>
Diagnostics					
R^2	11.2%	31.9%	22.9%	44.8%	31.2%
Wald test: all coeff.=0	11.4***	8.2***	6.4***	n.a.	n.a.
No. of firms	8,867	4,750	6,440	2,278	532
No. of observations	73,832	10,180	25,071	12,822	1,520

- Total leverage responds less than long-term leverage: consistent with adjustment occurring primarily in long-term debt.
- Market leverage responds less than book leverage: consistent with market values moving with equity prices, diluting mechanical leverage changes.
- **Extensive margin:** tax increases raise the probability of issuing long-term debt, while tax cuts do not raise the probability of repurchasing long-term debt.

8.6 Table 6: local business-cycle confounds and border designs

Threat: tax changes might be responses to local economic shocks that also affect leverage demand.

Strategies in the table:

1. **Neighbor-state falsification:** if the confound is a local shock spilling across borders, firms in neighboring states should “react” too. They do not in the same direction.
2. **Adjacent-state controls:** restricting controls to neighboring states (more comparable local conditions) yields even larger effects for tax increases and still no effects for tax cuts.

Table 7

Potential confounds: other tax changes, union power, and unemployment risk. This table investigates if our results are confounded by state-level variation in non-corporate taxes, union power, or unemployment risk. To address the first set of confounds, we control for concurrent changes in other taxes or tax credits (column 1). To address the next two confounds, we partition the sample into firms headquartered in states with high or low union power (columns 2 and 3) and those in states suffering large or no large employment shocks (columns 4 and 5). For variable definitions and details of their construction, see [Appendix C](#). For a description of the sample, see [Table 2](#). The unit of analysis is a firm-year. All specifications are estimated using OLS in first differences with industry-year fixed effects (not shown for brevity). Heteroskedasticity-consistent standard errors clustered at the state level are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 1%, 5%, and 10% level (two-sided), respectively. (Reflecting the signed nature of the predictions, the test for equal tax sensitivity is one-sided.)

	Dep. var.: Change in long-term book leverage				
	Baseline w/concurrent tax changes	States with ...		States with ...	
		High union power (2)	Low union power (3)	High mass layoffs (4)	Low mass layoffs (5)
=1 if tax increase at $t=-1$	1.084*** <i>0.195</i>	0.827*** <i>0.227</i>	1.467*** <i>0.529</i>	1.022*** <i>0.379</i>	1.042** <i>0.418</i>
=1 if tax cut at $t=-1$	-0.008 <i>0.143</i>	0.044 <i>0.223</i>	0.189 <i>0.316</i>	0.210 <i>0.255</i>	0.357 <i>0.253</i>
Lagged change in ...					
ROA	-0.005 <i>0.004</i>	-0.016*** <i>0.004</i>	0.003 <i>0.009</i>	-0.003 <i>0.004</i>	0.000 <i>0.007</i>
Firm size	0.007*** <i>0.002</i>	0.006*** <i>0.002</i>	0.007 <i>0.004</i>	0.007*** <i>0.001</i>	0.010*** <i>0.003</i>
Tangibility	0.037*** <i>0.011</i>	0.020 <i>0.013</i>	0.037* <i>0.021</i>	0.053*** <i>0.011</i>	0.037 <i>0.024</i>
Market/book	0.000 <i>0.000</i>	0.000 <i>0.001</i>	-0.001 <i>0.001</i>	0.000 <i>0.000</i>	-0.001 <i>0.001</i>
Default spread	-0.510*** <i>0.123</i>	-0.353* <i>0.196</i>	-0.087 <i>0.476</i>	-0.381** <i>0.161</i>	-0.589*** <i>0.184</i>
GSP growth rate	0.026 <i>0.019</i>	0.040 <i>0.030</i>	0.034 <i>0.048</i>	-0.057 <i>0.041</i>	0.014 <i>0.029</i>
State unemployment rate	0.178** <i>0.085</i>	0.025 <i>0.119</i>	0.227 <i>0.188</i>	-0.153 <i>0.194</i>	0.185 <i>0.143</i>
State tax on personal income	-0.050 <i>0.105</i>				
State tax on capital gains	0.011 <i>0.089</i>				
State tax on banks	-0.058 <i>0.153</i>				
State investment tax credits	0.001 <i>0.027</i>				
State R&D tax credits	-0.027 <i>0.023</i>				
State job tax credits	0.002 <i>0.002</i>				
Diagnostics					
R ²	11.2%	18.8%	28.4%	16.9%	20.3%
Wald test: all coeff.=0	14.4***	8.2***	8.7***	7.3***	7.2***
Equal tax sensitivity?	n.m.		0.86		0.01
No. of firms	8,867	5,424	2,859	5,215	3,852
No. of observations	73,832	37,507	19,039	28,560	20,488

3. **Contiguous border counties:** comparing firms headquartered in counties on either side of a state border and absorbing county-pair/year fixed effects isolates sharp variation in tax policy while holding local economic conditions fixed. Estimated effects remain positive and large.

8.7 Table 7: other policy confounds, union power, and unemployment risk

Threat: corporate tax changes might coincide with other tax changes or policies affecting leverage.

Key messages:

- Adding controls for concurrent changes in other state taxes and investment credits does not change the tax-increase effect.
- Partitioning by union power (motivated by a strategic leverage channel) does not support the view that union power drives the effect; if anything, the tax effect is larger in low-union states.
- Partitioning by unemployment-risk proxies yields similar tax-increase sensitivity in both groups.

Table 8

Heterogeneous treatment effects. Higher taxes on equity income (τ_e) dampen the impact of corporate tax changes on leverage. To test this, columns 1 and 2 split the sample according to a proxy for τ_e : non-dividend payers have lower τ_e than dividend-payers because their investors derive their equity income solely in the form of (lower-taxed) capital gains. A corollary of a causal interpretation of the observed tax sensitivity of debt is that it should vary with profits. Columns 3 and 4 partition sample firms according to whether they are profitable or loss-making in year 0. Columns 5–7 use John Graham’s simulated marginal tax rates to partition sample firms into those with marginal tax rates that are zero, positive, or in the top decile, respectively. The extent to which a firm can borrow more when faced with a tax rise depends on its debt capacity. Columns 8 and 9 partition firms into those rated investment-grade or below-investment-grade by a credit rating agency. For variable definitions and details of their construction, see Appendix C. For a description of the sample, see Table 2. The unit of analysis is a firm-year. All specifications are estimated using OLS in first differences with industry-year fixed effects (not shown for brevity). Heteroskedasticity-consistent standard errors clustered at the state level are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 1%, 5%, and 10% level (two-sided), respectively. (Reflecting the signed nature of the predictions, the test for equal tax sensitivity is one-sided.)

	Dep. var.: Change in long-term book leverage								
	Non-dividend payers (1)	Dividend payers (2)	Profitable (3)	Loss- making (4)	Firms w/zero marginal tax rates (5)	Firms w/positive marginal tax rates (6)	Top decile of marginal tax rates (7)	Investment grade (8)	Below invest- ment grade (9)
= 1 if tax rise at $t=-1$ (in %)	1.433*** 0.276	0.044 0.282	1.093*** 0.200	0.288 0.573	0.688 1.071	1.067*** 0.174	2.517** 1.056	0.887** 0.434	0.477 1.501
= 1 if tax cut at $t=-1$ (in %)	-0.028 0.180	0.102 0.188	-0.016 0.165	-0.124 0.440	-0.010 0.413	0.106 0.131	-0.226 0.512	0.101 0.304	-0.004 0.715
Lagged change in ...									
ROA	-0.007* 0.004	0.011 0.012	-0.008* 0.004	0.004 0.006	0.005 0.008	-0.006* 0.004	-0.073*** 0.022	-0.018 0.026	0.019 0.046
Firm size	0.008*** 0.002	-0.003 0.004	0.008*** 0.002	0.009** 0.004	0.013 0.007	0.007*** 0.002	0.012* 0.007	-0.002 0.007	0.020** 0.009
Tangibility	0.036*** 0.012	0.057** 0.024	0.025* 0.013	0.067*** 0.018	0.044* 0.024	0.030*** 0.011	0.094*** 0.031	0.038* 0.023	0.063 0.045
Market/book	0.000 0.000	-0.004** 0.001	-0.001 0.001	0.000 0.000	0.001 0.001	-0.001** 0.0004	0.002 0.001	-0.004** 0.002	-0.003 0.004
Default spread	-0.494*** 0.219	-0.658*** 0.245	-0.625*** 0.094	-0.312 0.401	-1.059 2.488	-0.496*** 0.117	-1.654*** 0.897	-0.080 0.309	-1.579*** 0.745
GSP growth rate	0.039 0.025	0.026 0.037	0.025 0.021	0.018 0.048	-0.045 0.118	-0.023 0.018	0.052 0.084	-0.005 0.039	0.025 0.096
State unemployment rate	0.286*** 0.111	-0.070 0.169	0.221*** 0.081	-0.042 0.207	0.198 0.406	0.037 0.100	0.112 0.495	-0.062 0.206	0.179 0.469
Diagnostics									
R ²	14.6%	32.3%	14.4%	26.8%	30.5%	12.2%	51.0%	11.3%	42.7%
Wald test: all coeff.=0 (F)	14.1***	5.6***	15.2***	3.2***	3.0***	14.3***	6.4***	8.8***	1.7*
Equal tax sensitivity? (F)		10.45***		1.60		0.08	n.a.		0.21
No. of firms	7,841	2,392	7,230	4,588	3,391	8,295	2,482	672	1,792
No. of observations	52,490	21,021	58,017	15,815	6,570	64,995	6,968	6,634	10,728

8.8 Table 8: heterogeneous treatment effects as validation tests

This table is designed as a *mechanism validation* for a causal tax-shield interpretation.

Predictions and findings:

- **Equity-income taxes (τ_e):** firms proxying for lower τ_e (non-dividend payers) respond more strongly to tax increases than dividend payers.
- **Profits:** profitable firms respond strongly; loss-making firms respond weakly (consistent with needing taxable income to use the marginal tax shield).
- **Marginal tax rates:** firms with higher simulated marginal tax rates react more strongly.
- **Debt capacity:** investment-grade firms respond; riskier firms do not, consistent with tighter distress constraints.

Across all splits, the response to tax *cuts* remains close to zero.

9 Economic magnitude: translating coefficients into “how big”

A useful way to summarize the magnitude is an elasticity:

$$\text{Elasticity} \equiv \frac{\partial L/L}{\partial \tau_c/\tau_c}, \quad (15)$$

where L denotes leverage and τ_c the corporate tax rate. Using the continuous-tax specification, the paper reports an elasticity around 0.5 for tax increases and close to zero for tax cuts. The broad interpretation is that marginal costs of debt are relatively flat for the average firm at observed leverage levels, so tax-induced changes in the tax shield can move leverage meaningfully.

10 Bottom line takeaways

1. **Causal evidence:** state corporate tax increases raise firms' long-term leverage with a one-year lag; tax cuts do not induce symmetric deleveraging.
2. **Mechanism:** the asymmetry and persistence are consistent with dynamic tradeoff models (ratcheting / hysteresis) rather than a frictionless static tradeoff.
3. **Methodological template:** combine a transparent quasi-experimental shock with multiple "threat-to-identification" checks (pre-trends, falsification, border designs, confound controls, and heterogeneity validation).

11 Conclusion

11.1 Summary

This paper concludes that **corporate taxes are a first-order determinant of leverage**, but the leverage response is **highly asymmetric**. Using a natural experiment based on **121 staggered changes** in U.S. *state* corporate income tax rates, the authors estimate how firms adjust long-term leverage when the tax benefit of debt rises or falls.

The paper's empirical and conceptual conclusions can be stated as a tight chain:

- (i) **Causal tax sensitivity exists and is economically meaningful.** Following a state corporate tax *increase*, firms raise long-term book leverage by about **1 percentage point** (one year later, in the baseline specification). In continuous-tax specifications, a **1 percentage-point** rise in the corporate tax rate increases long-term book leverage by roughly **0.39 percentage points** (about **40 bps**), consistent with the paper's abstract summary.
- (ii) **The response is asymmetric: tax cuts do not trigger symmetric deleveraging.** When state corporate tax rates *fall*, firms do *not* reduce leverage in a mirror-image way; estimated effects for tax cuts are near zero across specifications.
- (iii) **Tax increases "ratchet" leverage upward (hysteresis).** Even when a tax increase is later reversed by a subsequent tax cut, leverage does not fall back: the earlier leverage increase appears **persistent**. This indicates **stickiness on the downside** rather than frictionless mean reversion toward a static target.
- (iv) **The patterns validate a tax channel rather than a generic local-shock story.** Heterogeneity tests show stronger tax sensitivity exactly where the tax-shield mechanism predicts:
 - **Profitable** firms (with taxable income) respond more than loss-making firms.
 - Firms with **higher marginal tax rates** respond more than those with lower marginal rates.
 - **Investment-grade** firms respond more than junk-rated firms (consistent with debt capacity / distress constraints).

- (v) **Robustness supports a causal interpretation.** The results survive extensive checks: timing/event-study evidence (no meaningful pre-trends), border-county designs and neighbor-state falsifications (local-cycle confounds), controls for other coincident state-level policy changes, and corrections for headquarters-location mismeasurement (which otherwise attenuates the estimated effect).
- (vi) **Theory implication: the data reject a purely static tradeoff comparative statics.** A frictionless static tradeoff model predicts that leverage reacts *symmetrically* to tax increases and tax cuts. The observed asymmetry and persistence instead support **dynamic tradeoff models** in which leverage can increase when incentives strengthen but does not costlessly unwind when incentives weaken.
- (vii) **External-validity caveat.** The identified tax changes are relatively small (a few percentage points at most), so the estimates should not be mechanically extrapolated to very large tax reforms.

11.2 Why leverage responds to tax increases but not to tax cuts

A useful way to internalize the paper’s main result is to distinguish the **target** leverage from the **adjustment technology**.

1) Target leverage moves with the corporate tax rate. Let $L_t^*(\tau_{c,t})$ denote the desired leverage implied by the tax benefit of debt and the marginal cost of distress. A higher corporate tax rate $\tau_{c,t}$ raises the marginal value of interest deductibility, increasing L_t^* :

$$\frac{\partial L_t^*}{\partial \tau_{c,t}} > 0.$$

2) Adjustment is asymmetric because increasing vs. decreasing leverage uses different margins. Levering up can be done via new long-term borrowing (often when refinancing windows open). Delevering requires retiring debt, renegotiating, issuing equity, or shrinking the balance sheet—actions that can be costly due to issuance costs, refinancing frictions, covenants, agency conflicts, and the lumpy timing of capital-structure events.

3) A simple “ratchet” partial-adjustment representation. One reduced-form way to summarize the dynamic implication is:

$$L_t - L_{t-1} = \lambda^+ \max\{0, L_t^* - L_{t-1}\} + \lambda^- \min\{0, L_t^* - L_{t-1}\} + \varepsilon_t, \quad \lambda^+ > \lambda^- \approx 0. \quad (16)$$

If λ^+ is meaningfully positive but λ^- is close to zero, then:

- a tax *increase* raises L_t^* and produces a positive adjustment in L_t ,
- a tax *cut* lowers L_t^* , but the negative adjustment is weak, so leverage does not fall much.

This exactly matches the paper’s empirical asymmetry and the persistence after reversals.

4) **Why heterogeneity is so informative.** The paper’s cross-sectional patterns line up with the mechanism in (16):

- **Profitability:** if taxable income is low, the marginal tax shield is low, so L_t^* moves less and/or the incentive to issue debt is weaker.
- **Debt capacity:** higher default risk makes the marginal cost of debt steep; even if L_t^* increases, firms may be constrained from borrowing more.

Thus, stronger responses among profitable and investment-grade firms are a direct “mechanism check” for the tax channel.

11.3 Takeaway

State corporate tax hikes cause firms to lever up (about 1 pp in long-term book leverage; about 40 bps per 1 pp tax-rate increase), but tax cuts do not cause symmetric deleveraging, implying leverage ratcheting/hysteresis consistent with dynamic (not static) tradeoff models.